$\boxed{1.}$ To find the generators of the Lorentz group that leave k^μ invariant, we have the following condition on the infinitesimal transformation of k^μ

$$0 \overset{!}{=} \delta k_\mu = \omega_{\mu\nu} k^\nu \implies \omega_{i0} + \omega_{i3} = 0 \text{ and } \omega_{03} = 0 = \omega_{30}.$$

Using these conditions we obtain

$$\begin{aligned}
\omega_{\mu\nu} &= \frac{i}{2}\omega_{\alpha\beta}(M_{\mu\nu})^{\alpha\beta} \\
&= i\omega_{01}(M_{\mu\nu})^{01} + i\omega_{02}(M_{\mu\nu})^{02} + i\omega_{03}(M_{\mu\nu})^{03} + i\omega_{12}(M_{\mu\nu})^{12} + i\omega_{31}(M_{\mu\nu})^{31} + i\omega_{23}(M_{\mu\nu})^{23} \\
&= i\omega_{12}(M_{\mu\nu})^{12} + i\omega_{23}\left[(M_{\mu\nu})^{23} - (M_{\mu\nu})^{20}\right] + i\omega_{13}\left[(M_{\mu\nu})^{13} - (M_{\mu\nu})^{10}\right] \\
&= -i\omega_{12}(J^3)_{\mu\nu} - i\omega_{23}\left[(J^1)_{\mu\nu} + (N^2)_{\mu\nu}\right] - i\omega_{13}\left[-(J^2)_{\mu\nu} + (N^1)_{\mu\nu}\right].
\end{aligned}$$

Here, we can clearly see that the group which leaves k^μ invariant is three-parametric with the following generators:

$$J^3, \qquad T^1 = J^1 + N^2, \qquad T^2 = -J^2 + N^1.$$

Using the standard commutation relations of J^i, N^i generators

$$[J^i, J^j] = i\varepsilon^{ij}{}_k J^k, \qquad [N^i, N^j] = -i\varepsilon^{ij}{}_k J^k, \qquad [J^i, N^j] = i\varepsilon^{ij}{}_k N^k,$$

we obtain the commutation relations of the little group

$$\begin{aligned}
[J^3, T^1] &= [J^3, J^1 + N^2] = iJ^2 - iN^1 = -iT^2 \\
[J^3, T^2] &= [J^3, -J^2 + N^1] = iJ^1 + iN^1 = iT^1 \\
[T^1, T^2] &= [J^1 + N^2, -J^2 + N^1] = -iJ^3 + iJ^3 = 0,
\end{aligned}$$

which satisfy the standard commutation relations of the $\mathfrak{so}(2) \ltimes \mathbb{R}^2$ group.

$\boxed{2.}$ We need to solve the equation

$$D(J^3)\cdot\varepsilon_\pm = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \pm\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix},$$

from which we immediately get

$$\varepsilon^\mu_\pm = \mathcal{N}_\pm\left(\boldsymbol{e}_1 \pm i\boldsymbol{e}_2\right).$$

The scalar products of these vectors are

$$\begin{aligned}
\varepsilon_+ \cdot \varepsilon_+ &= -\mathcal{N}_+^2 (1-1) = 0 \\
\varepsilon_- \cdot \varepsilon_- &= -\mathcal{N}_-^2 (1-1) = 0 \\
\varepsilon_+ \cdot \varepsilon_- &= -\mathcal{N}_+\mathcal{N}_- (1+1) = -2\mathcal{N}_+\mathcal{N}_-
\end{aligned}$$

To obtain the desired normalization

$$\varepsilon_+ \cdot \varepsilon_- = -1,$$

we symmetrically choose $\mathcal{N}_+ = -\mathcal{N}_- = \frac{1}{\sqrt{2}}$.

$\boxed{3.}$ For the T^i generators, we have

$$D(T^1)\cdot\varepsilon_\pm = \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix} \frac{\mp 1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix} = \frac{-1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \neq 0$$

$$D(T^2)\cdot\varepsilon_\pm = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & -i \\ 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix} \frac{\mp 1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix} = \frac{\mp 1}{\sqrt{2}} \begin{pmatrix} i \\ 0 \\ 0 \\ i \end{pmatrix} \neq 0$$

$\boxed{5.}$ Using the hint, we have

$$\varepsilon_\pm(\Lambda p) = \Lambda L(p) W^{-1}(\Lambda, p)\varepsilon_\pm(k) = \Lambda L(p) e^{-i\vartheta(p,\Lambda)J^3} e^{-ib^j(p,\Lambda)T_j}\varepsilon_\pm(k).$$

Now we use the results of the 2nd and 3rd exercise. We know that

$$J^3\cdot\varepsilon_\pm(k) = \pm\varepsilon_\pm(k)$$

and also

$$T^1\cdot\varepsilon_\pm(k) = \frac{1}{\sqrt{2}}(1,0,0,1)^{\mathrm{T}} = \frac{-1}{\sqrt{2}}\frac{k}{E}, \qquad T^2\cdot\varepsilon_\pm(k) = \frac{\mp i}{\sqrt{2}}(1,0,0,1)^{\mathrm{T}} = \frac{\mp i}{\sqrt{2}}\frac{k}{E}.$$

Here we see that $T^j\varepsilon_\pm \propto k$, and because we have defined T^j as generators of a group which leaves k invariant, it must hold that $T^j\cdot k = 0$. Therefore, we conclude that the Taylor expansion of the exponential vanishes in higher than linear terms and we obtain

$$e^{-i(b^1T_1+b^2T_2)}\varepsilon_\pm(k) = \varepsilon_\pm(k) - i(b^1 \mp ib_2)\frac{k}{\sqrt{2}E} + 0 + \cdots .$$

The exponentiation of J^3 is now straightforward since $\varepsilon_\pm$ are its eigenvectors. Overall, we obtain:

$$\varepsilon_\pm(\Lambda p) = \Lambda L(p) e^{\mp i\vartheta(p,\Lambda)} \left[\varepsilon_\pm(k) - i(b^1 \mp i b_2)\frac{k}{E}\right].$$

Now, we just apply the canonical boost to obtain

$$\varepsilon_\pm(\Lambda p) = e^{\mp i\vartheta(p,\Lambda)} \Lambda \cdot \varepsilon_\pm(p) - i(b^1 \mp i b_2)\frac{\Lambda \cdot p}{\sqrt{2}E} e^{\mp i\vartheta(p,\Lambda)}.$$

In this relation we clearly see that the first part transforms Lorentz-covariantly, while the second part spoils the covariance.

$\boxed{6.}$ Firstly, we use the definition of the field

$$U(\Lambda)A^\mu(x)U^\dagger(\Lambda) = \sum_{h=\pm}\int \mathrm{d}\tilde{p}\, \left[U(\Lambda)a(p,h)U^\dagger(\Lambda)\varepsilon^\mu_h(p)e^{-ip\cdot x} + \text{h.c.}\right].$$

Now we use the relation for the transformation of the creation/annihilation operators

$$U(\Lambda)a(p,h)U^\dagger(\Lambda) = a(\Lambda p, h)e^{-ih\vartheta(p,\Lambda)},$$

and therefore we have

$$\begin{aligned} U(\Lambda)A^\mu(x)U^\dagger(\Lambda) &= \sum_{h=\pm}\int \mathrm{d}\tilde{p}\, \left[a(\Lambda p,h)e^{-ih\vartheta(p,\Lambda)}\varepsilon^\mu_h(p)e^{-ip\cdot x} + \text{h.c.}\right] \\ &= \sum_{h=\pm}\int \mathrm{d}\tilde{p}\, \left[a(p,h)e^{-ih\vartheta(p,\Lambda)}\varepsilon^\mu_h(\Lambda^{-1}p)e^{-ip\cdot\Lambda x} + \text{h.c.}\right], \end{aligned}$$

where we have used the Lorentz invariance of the $\mathrm{d}\tilde{p}$ measure. Now we can use the relation derived in the previous task:

$$\begin{aligned} U(\Lambda)A(x)U^\dagger(\Lambda) = \sum_{h=\pm}\int \mathrm{d}\tilde{p}\, \Big[&a(p,h)e^{-ih\vartheta(p,\Lambda)}e^{-ih\vartheta(p,\Lambda^{-1})} \\ &\left(\Lambda^{-1}\cdot\varepsilon_\pm(p) - i(b^1 \mp i b_2)\frac{\Lambda^{-1}\cdot p}{\sqrt{2}E}\right)e^{-ip\cdot\Lambda x} + \text{h.c.}\Big] \\ = \Lambda^{-1}\cdot A(\Lambda x) &- i\sum_{h=\pm}\int \mathrm{d}\tilde{p}\left[a(p,h)\frac{b^1 - ihb^2}{\sqrt{2}E}\left(\Lambda^{-1}\cdot p\right)e^{-ip\cdot\Lambda x} + \text{h.c.}\right] \\ = \Lambda^{-1}\cdot A(\Lambda x) &- i\sum_{h=\pm}\int \mathrm{d}\tilde{p}\left[a(p,h)\frac{b^1 - ihb^2}{\sqrt{2}E}\left(\Lambda^{-1}\cdot p\right)e^{-i\Lambda^{-1}p\cdot x} + \text{h.c.}\right] \end{aligned}$$

$$= \Lambda^{-1} \cdot A(\Lambda x) + \partial_x \sum_{h=\pm} \int \mathrm{d}\tilde{p} \left[a(p,h) \frac{b^1 - ihb^2}{\sqrt{2}E} e^{-i\Lambda^{-1}p \cdot x} + \text{h.c.} \right]$$

$$= \Lambda^{-1} \cdot A(\Lambda x) + \partial_x \sum_{h=\pm} \int \mathrm{d}\tilde{p} \left[a(p,h) \frac{b^1 - ihb^2}{\sqrt{2}E} e^{-ip \cdot \Lambda x} + \text{h.c.} \right]$$

Overall, we get the desired result

$$U(\Lambda) A^\mu(x) U^\dagger(\Lambda) = \Lambda_\nu{}^\mu A_\nu(\Lambda x) + \partial^\mu \alpha(\Lambda x),$$

where

$$\alpha(\Lambda x) = \sum_{h=\pm} \int \mathrm{d}\tilde{p} \left[a(p,h) \frac{b^1(p,\Lambda^{-1}) - ihb^2(p,\Lambda^{-1})}{\sqrt{2}E} e^{-ip \cdot \Lambda x} + \text{h.c.} \right].$$